SG No. 131 D_{4h}^{9} $P4_2/mmc$ [tetragonal]

* plus set: + [0, 0, 0]

Table 1: Wyckoff site: 2a, site symmetry: mmm.

No.	position	mapping
1	[0, 0, 0]	[1,2,5,6,9,10,13,14]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[3,4,7,8,11,12,15,16]

Table 2: Wyckoff site: 2b, site symmetry: mmm.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[1,2,5,6,9,10,13,14]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,4,7,8,11,12,15,16]

Table 3: Wyckoff site: 2c, site symmetry: mmm.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	[1,2,5,6,9,10,13,14]
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[3,4,7,8,11,12,15,16]

Table 4: Wyckoff site: 2d, site symmetry: mmm.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,2,5,6,9,10,13,14]
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	[3,4,7,8,11,12,15,16]

Table 5: Wyckoff site: 2e, site symmetry: -4m2

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,2,7,8,11,12,13,14]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[3,4,5,6,9,10,15,16]

Table 6: Wyckoff site: 2f, site symmetry: -4m2

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,2,7,8,11,12,13,14]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[3,4,5,6,9,10,15,16]

Table 7: Wyckoff site: 4g, site symmetry: 2mm.

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,13,14]

continued ...

Table 7

No.	position	mapping
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[3,4,15,16]`
3	$[0,\, 0,\, -z]$	`[5,6,9,10]`
4	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[7,8,11,12]`

Table 8: Wyckoff site: `4h`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, -z\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	`[7,8,11,12]`

Table 9: Wyckoff site: `4i`, site symmetry: `2mm`.

No.	position	mapping
1	$\left[0,\, \frac{1}{2},\, z\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[0,\, \frac{1}{2},\, -z\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-z\right]$	`[7,8,11,12]`

Table 10: Wyckoff site: `4j`, site symmetry: `m2m`.

No.	position	mapping
1	$[x,\, 0,\, 0]$	`[1,6,10,13]`
2	$[-x,\, 0,\, 0]$	`[2,5,9,14]`
3	$\left[0,\, x,\, \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[0,\, -x,\, \frac{1}{2}\right]$	`[4,8,11,15]`

Table 11: Wyckoff site: `4k`, site symmetry: `m2m`.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[1,6,10,13]`
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,5,9,14]`
3	$\left[\frac{1}{2},\, x,\, 0\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2},\, -x,\, 0\right]$	`[4,8,11,15]`

Table 12: Wyckoff site: `4l`, site symmetry: `m2m`.

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[1,6,10,13]`
2	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[2,5,9,14]`

continued ...

Table 12

No.	position	mapping
3	$[0,\, x,\, 0]$	`[3,7,12,16]`
4	$[0,\, -x,\, 0]$	`[4,8,11,15]`

Table 13: Wyckoff site: `4m`, site symmetry: `m2m`.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	`[1,6,10,13]`
2	$\left[-x,\, \frac{1}{2},\, 0\right]$	`[2,5,9,14]`
3	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[4,8,11,15]`

Table 14: Wyckoff site: `8n`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\, -x,\, \frac{1}{4}\right]$	`[2,8]`
3	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[3,5]`
4	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[4,6]`
5	$\left[-x,\, -x,\, \frac{3}{4}\right]$	`[9,15]`
6	$\left[x,\, x,\, \frac{3}{4}\right]$	`[10,16]`
7	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[11,13]`
8	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[12,14]`

Table 15: Wyckoff site: `8o`, site symmetry: `.m.`

No.	position	mapping
1	$[0,\, y,\, z]$	`[1,14]`
2	$[0,\, -y,\, z]$	`[2,13]`
3	$\left[-y,\, 0,\, z+\frac{1}{2}\right]$	`[3,15]`
4	$\left[y,\, 0,\, z+\frac{1}{2}\right]$	`[4,16]`
5	$[0,\, y,\, -z]$	`[5,10]`
6	$[0,\, -y,\, -z]$	`[6,9]`
7	$\left[y,\, 0,\, \frac{1}{2}-z\right]$	`[7,11]`
8	$\left[-y,\, 0,\, \frac{1}{2}-z\right]$	`[8,12]`

Table 16: Wyckoff site: `8p`, site symmetry: `.m.`

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	`[1,14]`
2	$\left[\frac{1}{2},\, -y,\, z\right]$	`[2,13]`
3	$\left[-y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[3,15]`
4	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[4,16]`

continued ...

Table 16

No.	position	mapping
5	$\left[\frac{1}{2}, y, -z\right]$	[5,10]
6	$\left[\frac{1}{2}, -y, -z\right]$	[6,9]
7	$\left[y, \frac{1}{2}, \frac{1}{2} - z\right]$	[7,11]
8	$\left[-y, \frac{1}{2}, \frac{1}{2} - z\right]$	[8,12]

Table 17: Wyckoff site: 8q, site symmetry: m..

No.	position	mapping
1	$[x, y, 0]$	[1,10]
2	$[-x, -y, 0]$	[2,9]
3	$\left[-y, x, \frac{1}{2}\right]$	[3,12]
4	$\left[y, -x, \frac{1}{2}\right]$	[4,11]
5	$[-x, y, 0]$	[5,14]
6	$[x, -y, 0]$	[6,13]
7	$\left[y, x, \frac{1}{2}\right]$	[7,16]
8	$\left[-y, -x, \frac{1}{2}\right]$	[8,15]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-x, -y, z]$	[2]
3	$\left[-y, x, z + \frac{1}{2}\right]$	[3]
4	$\left[y, -x, z + \frac{1}{2}\right]$	[4]
5	$[-x, y, -z]$	[5]
6	$[x, -y, -z]$	[6]
7	$\left[y, x, \frac{1}{2} - z\right]$	[7]
8	$\left[-y, -x, \frac{1}{2} - z\right]$	[8]
9	$[-x, -y, -z]$	[9]
10	$[x, y, -z]$	[10]
11	$\left[y, -x, \frac{1}{2} - z\right]$	[11]
12	$\left[-y, x, \frac{1}{2} - z\right]$	[12]
13	$[x, -y, z]$	[13]
14	$[-x, y, z]$	[14]
15	$\left[-y, -x, z + \frac{1}{2}\right]$	[15]
16	$\left[y, x, z + \frac{1}{2}\right]$	[16]